

Understanding and Reducing Gender Bias in AI Language Models

Alexey Kresin¹, Diefi Tchifou Miltiade¹, Tomer Caspi², Prof. Jason Miller¹
¹ Hood College, Frederick MD, United States ² Tel Aviv University, Tel Aviv, Israel

Background:

Language models process numbers, not words. Word embeddings represent each word as a vector in a high-dimensional space. Words with similar meanings are located close together, forming meaningful relationships. However, these relationships can also encode unwanted patterns, such as gender bias.

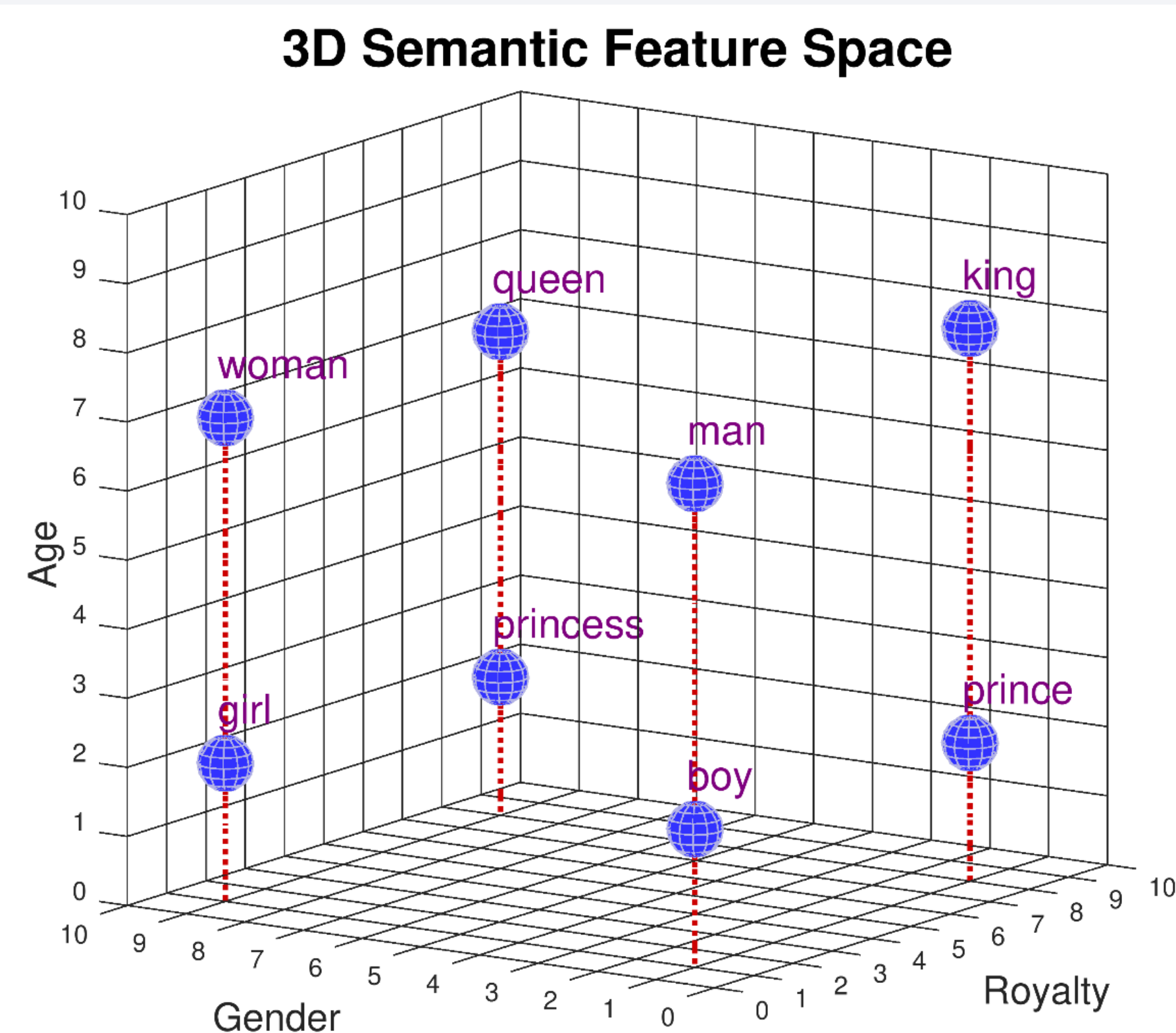


Fig 1. Illustration of embedding space where relationships between words are represented geometrically.

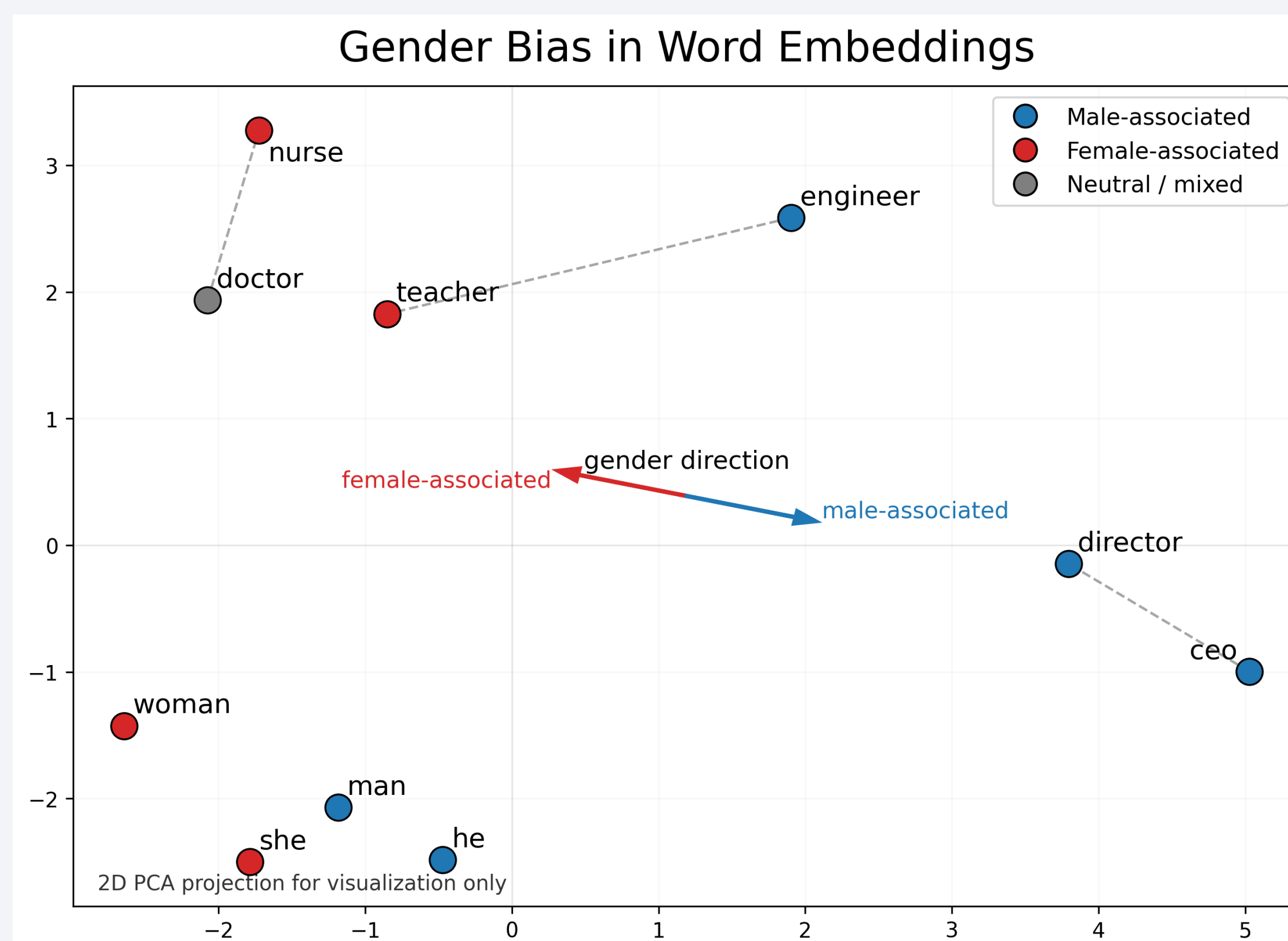


Fig 2. Example of gender bias in embeddings. Words align along a gender-related direction (she → he), and profession words are unevenly distributed (e.g., nurse vs engineer).

Prior Work & Limitation:

Prior work has shown that bias in embeddings can be reduced by removing a dominant gender direction.⁽¹⁾ However, recent studies suggest that bias may persist in more complex forms within the embedding space.⁽²⁾

Research Question:

How is gender bias distributed across embedding Components, and how do different metrics capture it?

Method:

We remove components identified by PCA and evaluate both bias reduction and semantic impact.

Step 1. Identify gender Components

We apply PCA to gender word pairs to identify components capturing gender information.

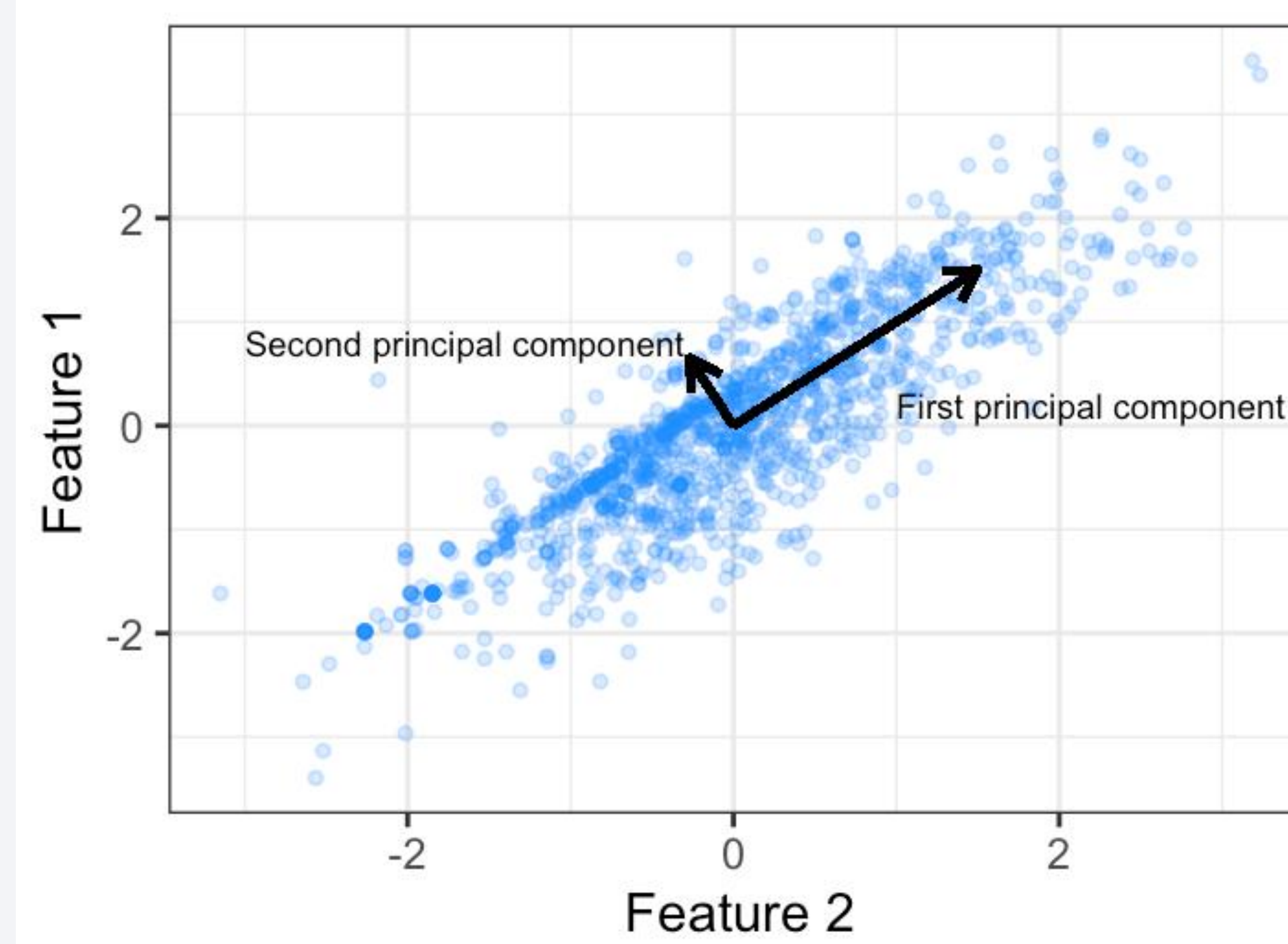


Fig 3. PCA illustration showing how components capture direction of variation

Step 2. Remove components

We progressively remove the top k components from the embedding

Step 3. Evaluate impact

We measure how bias and semantic structure change after removal

Evaluation Metrics:

Bias Reduction:

Direct Bias → alignment with gender component

WEAT → association between words and gender

Semantic Impact:

Displacement → how much word vectors change

Neighbor Stability → how much word meaning changes.

Results:

Bias and Geometry Respond Differently to Component Removal

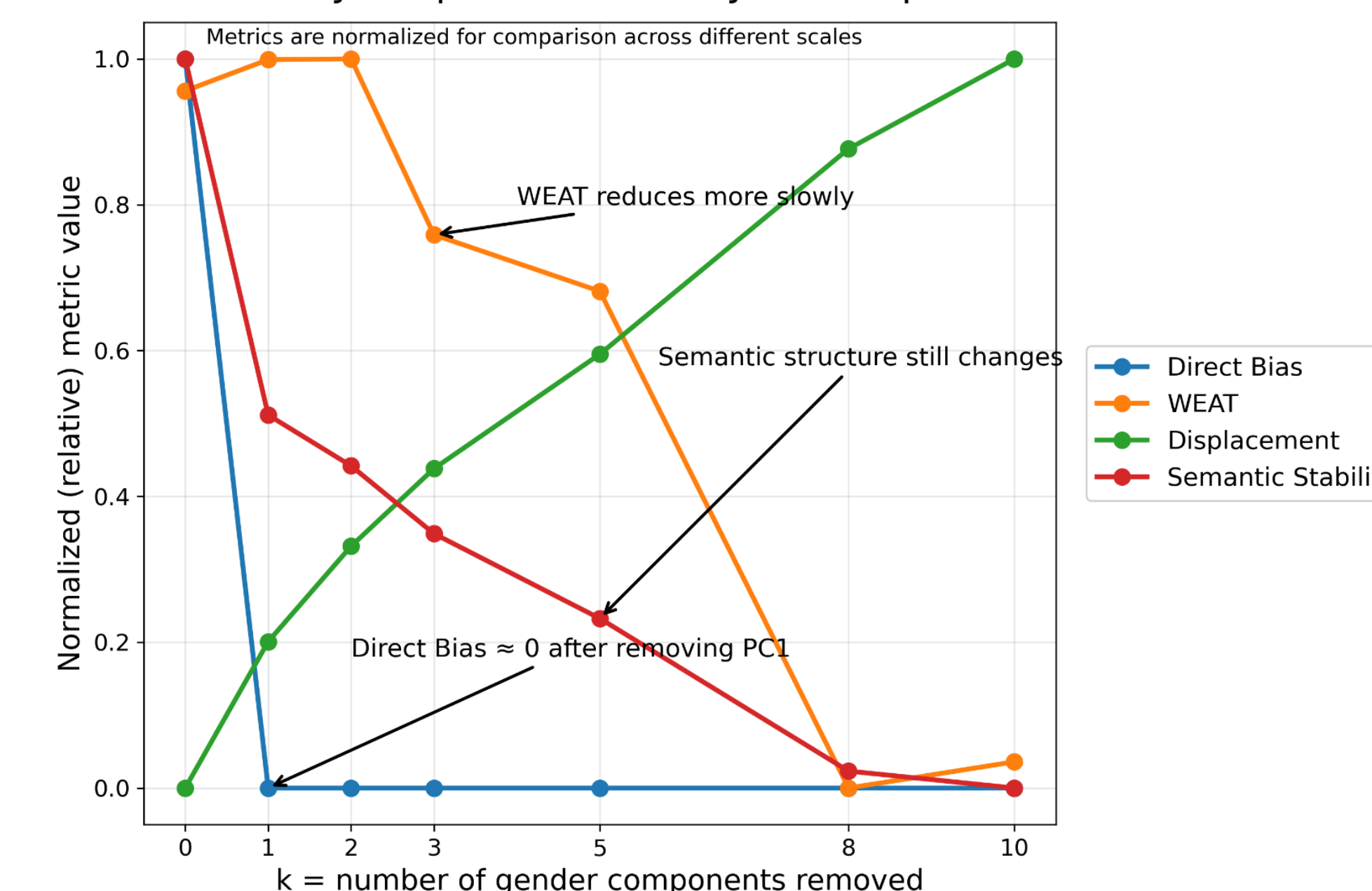


Fig 4. Bias and semantic metrics as components removed

Results Interpretation:

- Removing the first component (k=1) sharply reduces Direct Bias.
- However, WEAT (association bias) and semantic metrics continue to change as more components are removed.
- This shows that bias is not confined to a single component.
- Removing additional components reduces bias further but also distorts semantic relationships.

Key Findings:

- Removing a single gender component reduces
- Direct Bias but does not eliminate bias captured by other metrics
- Bias is distributed across multiple components in embedding space.
- Further removal reduces bias but increasingly distorts semantic meaning.
- Different metric capture different aspects of bias.

Next steps:

- Cross-validate results on multiple embedding models (word2vec, FastText)
- Develop a mathematical formulation of bias distribution in embedding space.
- Characterize which bias metrics capture specific aspects of bias and their relationship to embedding geometry.

References:

- [1] Bolukbasi, T., Chang, K., Zou, J., Saligrama, V., & Kalai, A. (2016). Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings. NeurIPS.
- [2] Caliskan, A., Bryson, J., & Narayanan, A. (2017). Semantics Derived Automatically from Language Corpora Contain Human-like Biases. Science.
- [3] Gonen, H., & Goldberg, Y. (2019). Lipstick on a Pig: Debiasing Methods Cover up Systematic Gender Biases in Word Embeddings. NAACL.

This is GitHub
to the project →

